

B K Birla Institute of Engineering & Technology, Pilani

45 Days Summer Training Program

Subject: Data Science with AIML

Name: Praveen Poonia

Id: 24EBKCS087

Branch: CS

College: BKBIET Pilani

Assignment 1

Que1. Explain the difference between the following data types with examples?

Python provides different data types to store different kinds of values. Some commonly used data types are Integer, Float, String, and Boolean.

(a) Integer (int) -

It is used to store whole numbers without any decimal point. Integers can be positive, negative, or zero.

Example:

```
age = 18  
marks = 95
```

(b)Float (float) -

float is used to store numbers that contain decimal values.

Example:

```
height = 5.8  
price = 99.99
```

(c)String (str) -

string is used to store text or a group of characters enclosed in quotation marks.

Example:

```
name = "Praveen Poonia"
```

(d) Boolean (bool) -

Boolean data type stores only two values: True or False. It is mainly used in decision-making and logical operations.

Example:

```
is_student = True
```

Que2. Write a Python program to create three variables: name, age, and city. Store your details in them and print the values?

Variables are used to store information in Python. The following program stores personal details and displays them.

```
name = "Praveen"  
age = 18  
city = "Pilani"
```

```
print("Name:", name)  
print("Age:", age)  
print("City:", city)
```

Output:

```
Name: Praveen Poonia  
Age: 18  
City: Pilani
```

Que3. Write a Python program that takes a user's name as input, prints the name in uppercase, and prints the total number of characters in the name?

This program accepts a name from the user, converts it into uppercase letters, and calculates the total number of characters present in the name.

```
name = input("Enter your name: ")

print("Uppercase Name:", name.upper())
print("Total Characters:", len(name))
```

Output:

```
Enter your name: Praveen Poonia
Uppercase Name: PRAVEEN POONIA
Total Characters: 14
```

Que4. Explain any five commonly used string methods in Python with examples?

String methods are built-in functions that help us perform various operations on strings.

1. upper()-

Converts all characters of a string into uppercase.

```
Example:
text = "python"
print(text.upper())
```

2. lower()-

Converts all characters of a string into lowercase.

```
Example:
text = "PYTHON"
print(text.lower())
```

3. replace()-

Replaces a word or character with another value.

```
Example :
text = "I like Java"
print(text.replace("Java", "Python"))
```

4. split()-

Splits a string into multiple parts and returns a list.

Example:

```
text = "Apple Mango Banana"  
print(text.split())
```

5. strip()-

Removes extra spaces from the beginning and end of a string.

Example:

```
text = " Hello "  
print(text.strip())
```

These methods make string manipulation easier and more efficient in Python.

Que5. Create a list containing the names of five fruits. Print the complete list, first element, last element, and total number of items?

A list is a collection used to store multiple values in a single variable.

Example:

```
fruits = ["Apple", "Mango", "Banana", "Orange", "Grapes"]
```

```
print("Complete List:", fruits)  
print("First Element:", fruits[0])  
print("Last Element:", fruits[-1])  
print("Total Items:", len(fruits))
```

Output:

```
Complete List: ['Apple', 'Mango', 'Banana', 'Orange', 'Grapes']  
First Element: Apple  
Last Element: Grapes  
Total Items: 5
```

Que6. Write a Python program to create a list of numbers [10, 20, 30, 40, 50], add 60 to the list, remove 20 from the list, and print the updated list?

Lists can be modified by adding or removing elements. The following program demonstrates these operations.

Example:

```
numbers = [10, 20, 30, 40, 50]
```

```
numbers.append(60)
```

```
numbers.remove(20)
```

```
print("Updated List:", numbers)
```

Output:

Updated List: [10, 30, 40, 50, 60]

Que7. What is Artificial Intelligence (AI)? Explain its importance and mention any four real-life applications of AI?

Artificial Intelligence (AI) is a branch of computer science that enables machines and software to perform tasks that normally require human intelligence. These tasks include learning, problem-solving, decision-making, understanding language, and recognizing patterns.

Importance of AI

- Improves efficiency and productivity.
- Reduces human effort in repetitive tasks.
- Helps in faster and better decision-making.
- Increases accuracy and automation.

Real-Life Applications of AI

1. ChatGPT and AI chatbots.
2. Voice assistants like Alexa and Siri.
3. Google Maps route and traffic prediction.
4. Netflix and YouTube recommendation systems.

AI is becoming an important part of daily life and is transforming many industries such as healthcare, education, transportation, and business.

Que8. Identify whether the following are examples of AI and explain why:

ChatGPT – AI

ChatGPT is an AI application because it can understand human language and generate meaningful responses using machine learning models.

Google Maps Route Prediction – AI

Google Maps uses AI to analyze traffic data, road conditions, and user behavior to suggest the fastest routes.

Calculator – Not AI

A calculator follows fixed mathematical instructions and does not learn from data or make decisions on its own.

Netflix Recommendations – AI

Netflix uses AI algorithms to study user preferences and recommend movies and shows that users may like.

Voice Assistants (Alexa/Siri) – AI

Voice assistants use speech recognition and natural language processing to understand and respond to voice commands.

Conclusion

Most modern smart applications use Artificial Intelligence to improve user experience and automate tasks. AI is one of the fastest-growing technologies in today's world.